MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

LOAD, RESISTOR, STANDARD ANALOG, KEY CODE FLB

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

1 SCOPE

1.1 Scope This document establishes the requirements and test procedures for the acquisition o an electronic standard resistor load module

1.2 Classification This specification covers one class of modules

Class 1 as specified in MIL-STD-1389

2 APPLICABLE DOCUMENTS

2.1 Government specifications and standards Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules
MIL-STD-1864 - Test Equipment for the Standard Electronic Module Program

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Naval Electronic Systems Command, ATTN ELEX 811, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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3 REQUIREMENTS

3 1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein

3 2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787

3 3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0 0120 failures per million operating hours when predicted in accordance with MIL-STD-1389

3 4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class 1 as specified in MIL-STD-1389

3 5 Module configuration requirements The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein

3 5 1 Key code The module key code shall be FLB

3 5 2 Size The module size shall be 1A

3 5 3 Weight The module shall have a maximum weight of 226 6 grams

3 5 4 Contact pins The module shall have 40 contact pins positioned A1 through A40

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A

3 5 6 Depth The module shall have no additional depth requirements

3 5 7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes

3 5 8 Marking Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/126-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier)

3 6 Circuit pin configuration The function diagram and pin assignments shall conform to figure 1

3 7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I

3 8 End-of-life limits Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6 2)

4 QUALITY ASSURANCE PROVISIONS

4 1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply

4 2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787

4 3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665 Other equipment and components shall be as specified on the figures All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified

4 4 Test procedures The module shall be tested as specified in table I

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NOTES
1. R1 = 10Ω
2. R2 = 20 5Ω
3. R3 = 30 1Ω
4. R4 = 40 2Ω
5. R5 = 100 kΩ
6. R6 = 205 kΩ
7. R7 = 301 kΩ
8. R8 = 402 kΩ
9. R9 = 10 kΩ
10. R10 = 20 5 kΩ
11. R11 = 30 1 kΩ
12. R12 = 40 2 kΩ
13. R13 = 1 kΩ
14. R14 = 2 05 k
15. R15 = 3 01 kΩ
16. R16 = 4 02 kΩ
17. R17 = 100Ω
18. R18 = 205Ω
19. R19 = 301Ω
20. R20 = 402Ω

All resistors - 1 4W

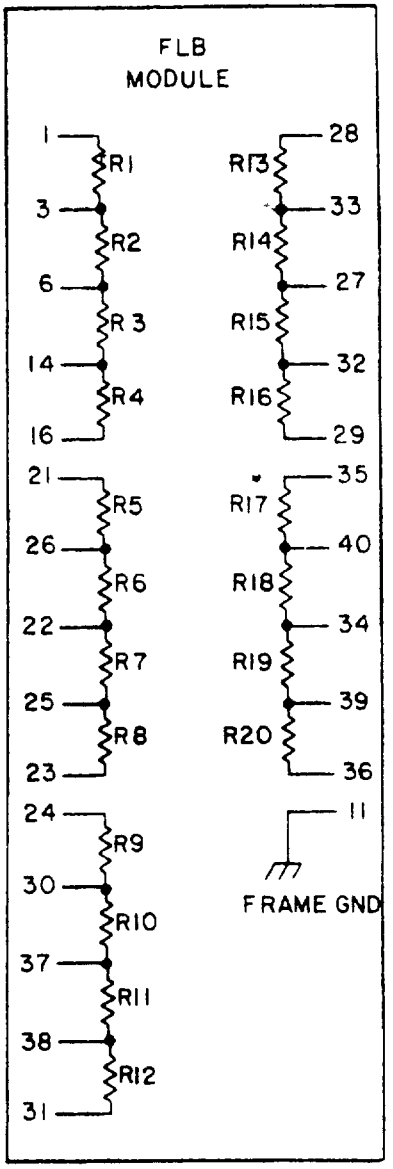


FIGURE 1. FLB module function diagram, standard resistor load

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TABLE I Module electrical requirements

Test condition	Parameter	Ref test table	Test type 1/, 2/	Acceptance requirements 25° +5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A5	Module integrity	II	X	10	-	10	-	--	--	--	--	Y
B1	R1 Resistance	III	X	9 9	10 1	9 475	10 525	9 825	10 175	9.3	10 7	Ω
B2	R2 Resistance	III	X	20 295	20 705	19 42	21 58	20 141	20 859	19 06	21.94	Ω
B3	R3 Resistance	III	X	29 799	30 401	28 52	31 68	29 573	30 627	27 99	32.21	Ω
B4	R4 Resistance	III	X	39 798	40 602	38.09	42 31	39 496	40 904	37.38	43.02	Ω
B5	R5 Resistance	III	X	99	101	94 75	105.25	98.25	101 75	93	107	kΩ
B6	R6 Resistance	III	X	202 95	207 05	194 2	115 8	201 41	208 59	190.6	219 4	kΩ
B7	R7 Resistance	III	X	297.99	34.01	285 2	316 8	295 75	306.25	279 9	322 1	kΩ
B8	R8 Resistance	III	X	397 98	406 02	380 9	423 1	394 96	409 04	373 8	430.2	kΩ
B9	R9 Resistance	III	X	9 9	10 1	9 475	10 525	9 825	10.175	9 3	10.7	kΩ
B10	R10 Resistance	III	X	20 295	20 705	19 42	21 58	20 141	20 859	19 06	21 94	k
B11	R11 Resistance	III	X	29 799	30 401	28 52	30 68	29 573	30 627	27 99	32 21	kΩ
B12	R12 Resistance	III	X	39 798	40 602	38 09	42 31	39 496	40 904	37 38	43 02	kΩ
B13	R13 Resistance	III	X	0 99	1 01	0 475	1 525	0 9825	1 0175	0 93	1 07	kΩ
B14	R14 Resistance	III	X	2 0295	2 0705	1 942	2 158	2 0141	2 0859	1.906	2 194	k

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TABLE I. Model electrical requirements - continued

Test condition	Parameter	Ref test table	Test type 1/, 2/	Acceptance requirements 25° +5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
B15	R15 Resistance	III	X	2.9799	3.0401	2.852	3.168	2.9573	3.0627	2.799	3.221	k
B16	R16 Resistance	III	X	3.9798	4.0602	3.809	4.231	3.9496	4.0904	3.738	4.302	k
B17	R17 Resistance	III	X	99	101	94.75	105.25	98.25	101.75	93	107	
B18	R18 Resistance	III	X	202.95	207.05	194.9	215.1	201.41	208.59	190.6	219.4	
B19	R19 Resistance	III	X	297.99	304.01	285.2	316.8	295.73	306.27	279.9	322.1	
B20	R20 Resistance	III	X	397.98	406.02	380.9	423.1	394.96	409.04	373.8	430.2	
C	Continuity	IV	S	0	1	0	1	--	--	--	--	

1/ X parameters shall be 100 percent tested during module acceptance tests
2/ S parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C

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4 4 1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787

4 4 2 Order of testing Test conditions A through C shall be performed in any order with the following exceptions If test condition A is part of a sequence of tests, the order of testing shall be as follows A where applicable, followed by the remaining tests in the sequence

4 5 Qualification testing Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions

4 5 1 Operating temperature tests The operating temperature tests shall be conducted in accordance with MIL-M-28787 Before starting these tests, test conditions A through C in table I shall be performed in the order specified in 4 4 2 using the initial limits at 25° ±5°C ambient When electrical testing is called for at the end of each temperature stabilization period, test condition B shall be performed in the order specified in 4 4 2 using the initial limits for applicable operating temperature The module shall be operated in the test circuit shown on figure 2 At all other times during the temperature tests, the module shall be inoperative

4 5 1 1 Component temperature test The component temperature test shall be conducted in accordance with MIL-M-28787 The module shall be operated as shown on figure 2

4.5 2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787

4 5 3 Life test The life test shall be conducted in accordance with MIL-M-28787 The module shall be operated as shown on figure 2 Power shall be applied intermittently 1 1/2 hours on and 1/2 hour off for the duration of the life test Following the life test and while still at the maximum class temperature, electrical tests A through C shall be performed in the order specified in 4 4 2 using the operating temperature end-of-life requirements of table I Following the return of the module to 25°C, electrical test B shall be performed in the order specified in 4 4 2 using the 25° ±5°C ambient end-of-life requirements of table I

5 PACKAGING

5 1 Packaging requirements The requirements for packaging shall be in accordance with MIL-M-28787

6 NOTES

6 1 Definitions Definitions are as specified in MIL-STD-1331

6 2 End-of-life When any individual module circuit parameter fails to conform to the end of life requirements of table I, the module is considered to have completed its useful life (see 3 6)

6 3 Cross reference of classification Military Specification MIL-M-28787/126 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No 10001-2658650 dated May 7 1971 Part number M28787/126-01 supersedes part numbers 2658650 and M28787/126 These parts are interchangeable

6 4 Module description The following electrical parameters reflect the 0 to 60 C end-of-life limits from table I

Maximum power dissipation	
Resistor	1/4 w
Module	1 1/4 W
Resistance loads	10, 20 5, 30 1, 40 2, 100, 205, 301, 402, and 1, 2 05, 3 01, 4 02, 10, 20 5, 30 1, 40 2, 100, 250, 301, 4 02

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6.5 Module function. The module consists of 20 resistors whose prime usage is for specific resistance loads.

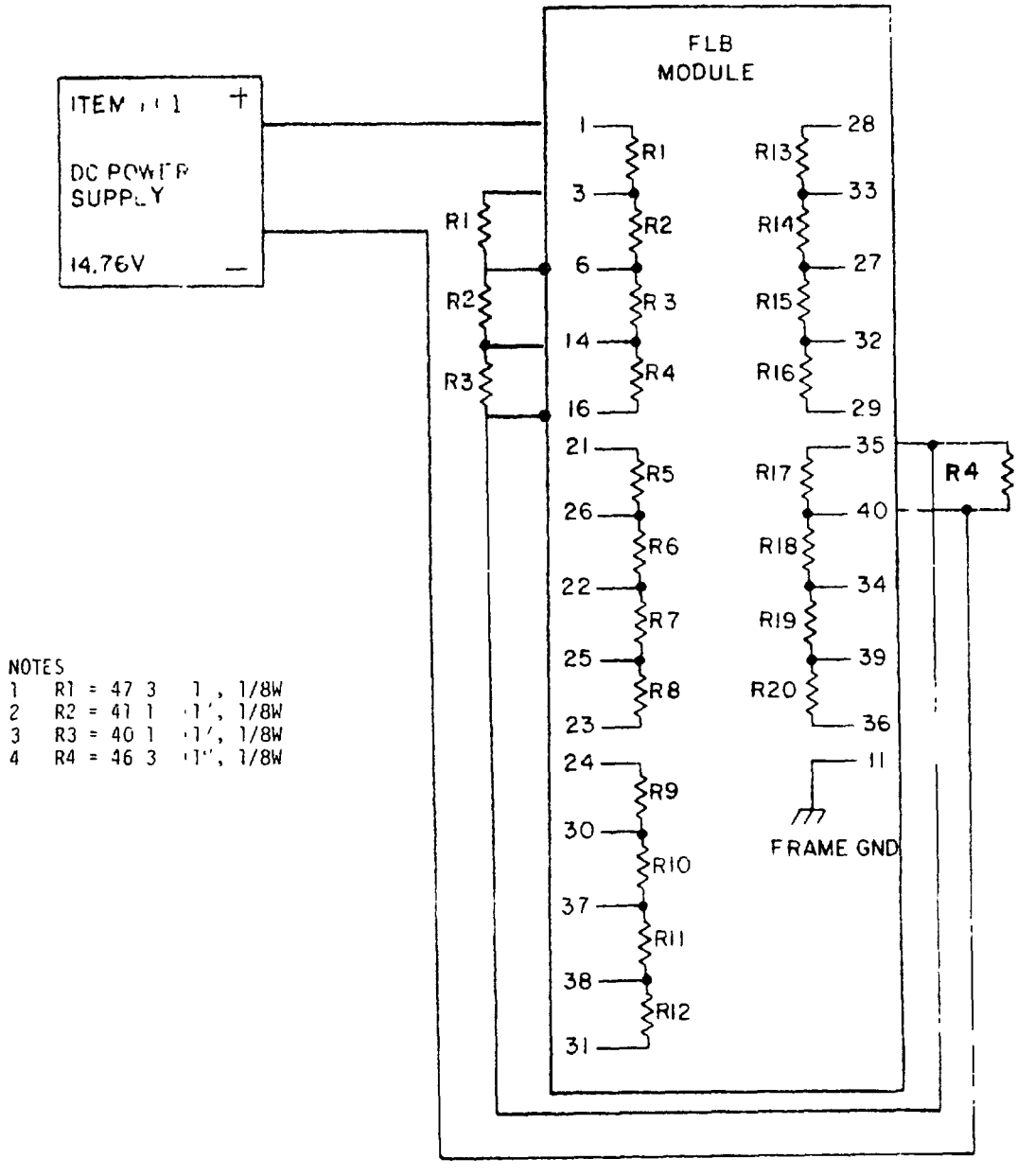
6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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NOTES
1. R1 = 47.3 Ω, 1/8W
2. R2 = 41.1 Ω, 1/8W
3. R3 = 40.1 Ω, 1/8W
4. R4 = 46.3 Ω, 1/8W

FIGURE 2. Operating and component temperature test and life test circuits.

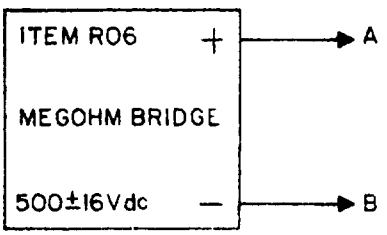
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TABLE II. Module integrity test

Test condition	Parameter	Terminal		Critical limits (-55°C Ambient)			
		A to pin	B to pin	Min	Max		
A1	Module integrity	1-8, 14-16, 21-40	9-13, 15, 17-20	10			
A2		1, 3, 6, 14-40	2, 4, 5, 7-13				
A3		1-6, 9, 10, 11, 14-18, 21-26, 30, 31, 37, 38	7, 8, 12, 13, 19, 20, 27-29, 32-36, 39, 40				
A4		5, 7, 11, 17, 18, 19, 24-30, 31, 34-40	1-4, 6, 8-10, 13-17, 19-23, 25-28, 32, 33				
A5		1, 2, 3, 6, 9, 14, 15, 16	4, 5, 7, 8, 10-13, 17-40				

MIL-M-28787/126A(USAF)

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WARNING

When using the megohm bridge, nigh voltage may be present at any of the output posts Always set the function switch of the megohm bridge to discharge before connecting or disconnecting the test leads

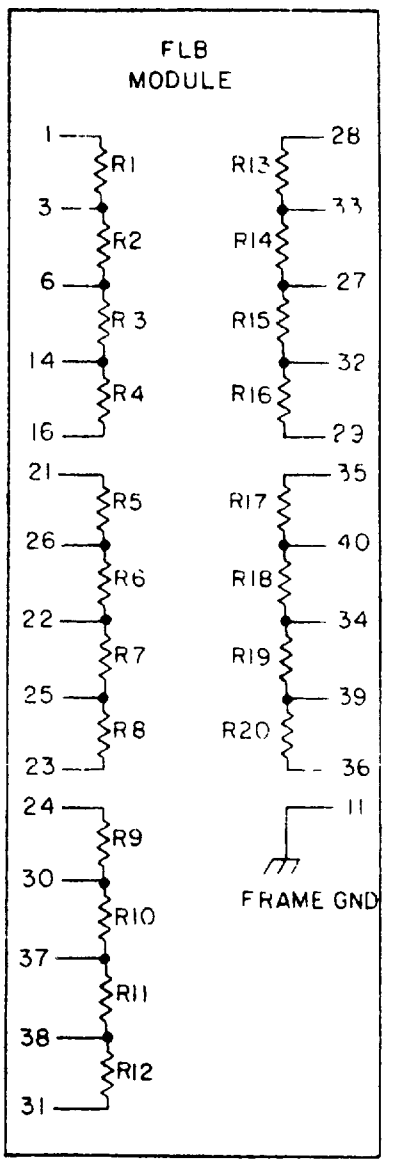


FIGURE 3 Module integrity test circuit for table II

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TABLE III. Resistance test

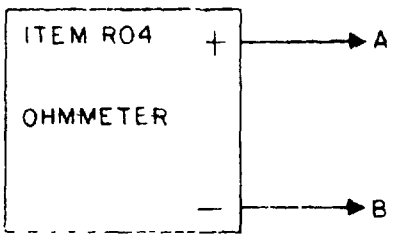
Test condition	Parameter	Contact		Initial limits -55°C, +25°C, +125°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	R1 Resistance	1	1	9.9	10.1	Ω	4
B2	R2	3	6	20.295	20.705	kΩ	
B3	R3	6	14	29.799	30.401		
B4	R4	14	16	39.798	40.602		
B5	R5	21	26	99	101	Ω	
B6	R6	26	22	202.95	207.05	kΩ	
B7	R7	22	25	297.99	34.01		
B8	R8	25	23	397.98	406.02		
B9	R9	24	30	9.9	10.1	Ω	
B10	R10	30	37	20.295	20.705	kΩ	
B11	R11	37	38	29.799	30.401		
B12	R12	38	31	39.798	40.602		
B13	R13	28	33	0.99	1.01	Ω	
B14	R14	33	27	2.0295	2.0705	kΩ	
B15	R15	27	32	2.9799	3.0401		
B16	R16	32	29	3.9798	4.0602		
B17	R17	35	40	99	101	Ω	
B18	R18	40	34	202.95	207.05	kΩ	
B19	R19	34	39	297.99	304.01		
B20	R20	39	36	397.98	406.02		

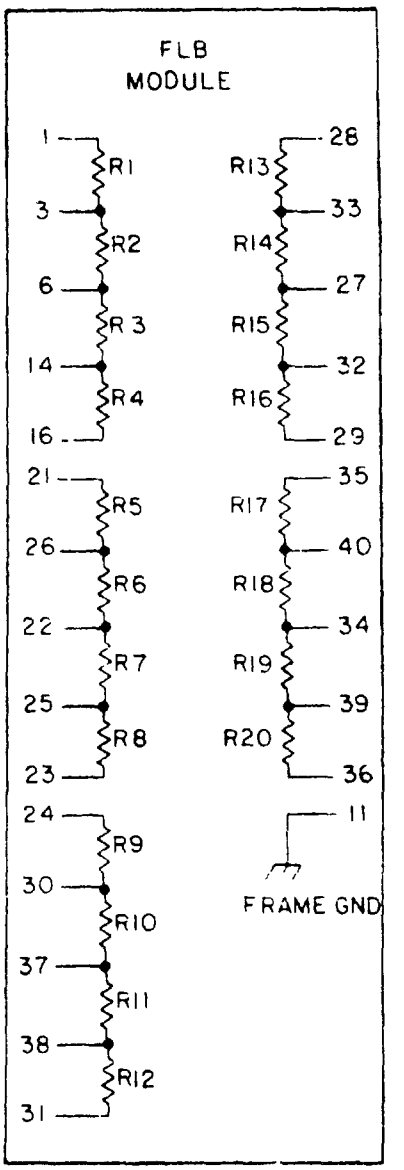
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TABLE IV Continuity test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C	Continuity	11	Frame	0	1	Ω	4

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NOTE: For test condition C, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal

PAGE - Resistance and continuity test - for tables I and II

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